Gregory Bian

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Education

University of British Columbia Expected May 2028
Bachelor of Applied Science, Computer Engineering *Vancouver, BC*

Technical Experience

Geotab Burnaby, BC
Software Engineer Intern – Vehicle Platform Development *May 2026 – Present*

- Developed production code on 6.5M+ vehicles, adding 15+ OEM vehicle types to Geotab's telematics platform
- Collected 15M+ data points on 25k+ vehicles by building signal workflows that debounce noise, fetch signal dependencies, and validate the data path to Google BigQuery with Python tests
- Migrated 100+ SIL test suites to use real, unmasked vehicle data for VIN decoding and vehicle identification, improving test coverage and removing the need for bench simulations
- Removed 50+ stale feature flags by automating the end-to-end removal process with Claude Skills on VMs, covering code deprecation, MR creation, CI/SIL monitoring, and Jira updates; adopted team-wide

UBC Solar Racing Vancouver, BC
Embedded Systems Team Lead *Jun 2026 – Present*

- Led a 9-person subteam within an 80+ member team, delivering firmware and telemetry for a solar-powered car
- Directed design and code reviews for 50+ features across firmware and telemetry, mentoring members on code standards, use of hardware and tooling, and system architecture
- Architected SIL and Regression testing pipelines using Renode hardware emulation and Robot Framework to simulate RTOS and hardware edge cases

Senior Designer *Apr 2026 – Jun 2026*

- Resolved 30+ firmware issues by analyzing 25M+ data points in Grafana and InfluxDB, unblocking electrical and strategy teams for driving days
- Built 10+ Python and Bash tools to simulate CAN traffic for bench tests and automate cellular and dev setup

Embedded Firmware Engineer *Sep 2025 – Apr 2026*

- Developed firmware to display 6x more data on the LCD Dashboard than previous iterations by architecting a multi-page design, improving real-time driver awareness, fault detection, and decision making
- Characterized and eliminated live telemetry packet loss (250 bytes/s $\rightarrow$ 0) by using RTOS semaphores in UART drivers for an XBee radio module to synchronize shared buffers across concurrent tasks
- Added 50+ vehicle signals to the CAN bus including motor temperature, GPS coordinates, and various diagnostics
- Diagnosed 5+ hardware issues on custom PCBs by probing signals with oscilloscopes, multimeters, and PCANs

Projects

Bare-Metal Tron Game | *C, RISC-V Assembly, FPGA* Nov 2025

- Developed a bare-metal Tron game in C, generating VGA output for real-time graphics rendering on a DE10-Lite FPGA and featuring multiplayer or single-player against a simple logic-driven robot.
- Implemented hardware interrupt–driven control logic using RISC-V Assembly to handle player and robot movement, collision detection, and real-time HEX display updates.

Extracurriculars

nwPlus Vancouver, BC
Engagement Coordinator *May 2026 – Present*

- Organized the largest hackathon in western Canada, hosting 750+ students, 50+ sponsors, and 150+ mentors
- Fostered a collaborative community within a 50+ member organizing committee by planning team-building events

Technical Skills

Programming: C, C++, Python, Java, JavaScript (React), Bash, ARM Assembly
Embedded Systems: FreeRTOS, STM32, CAN (OBD-II, UDS, ISO-TP, J1939), UART, SPI, I2C, SWD, JTAG
Tools: Linux, Git, CMake, J-Link, Renode, SavvyCAN, PCAN, Oscilloscope, Robot Framework, Grafana, InfluxDB